

APPROVED



BSc (Hons) Cyber Security
School of Computing, Engineering and Technology

Version	3.0
Course Type	Undergraduate
Award Type	Bachelor of Science with Honours - BSc (Hons)
Awards	
No Course Award Assigned	
Code	0531
SCQF Level	Level 10
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2026 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100365	computer networks	computer networks	100%
Total			100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The cyber security course employs a diverse array of teaching methods to accommodate various learning preferences. These include lectures, seminars, workshops, tutorials, computer laboratories, group exercises, case studies, oral presentations, practical work, an industrial placement, and a substantial project during the Honours year. The format chosen is tailored to each subject, ranging from staff-supported problem-solving to group exercises where responsibilities are distributed within each group. Teaching techniques are designed to discuss underlying theories, support peer learning, clarify material, and experiment with techniques and skills. Integration of online platforms, such as MentiMeter for quizzes, polls, word clouds, boards, and Padlet, or Office 365 tools for collaborative creations, enhances interactivity and immediacy in the learning process. A key emphasis is placed on cultivating the ability to work effectively within a project team, a crucial skill in the industry, making group work a significant aspect of the course at the Degree Stage. Practical experience with PCs or workstations is essential, and the School of Computing provides nine laboratories equipped with the latest software versions exclusively for student use.
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What the Course Involves	The School is equipped with two specialized labs dedicated to instructing cyber security modules. Each student is granted access to a "Pod," consisting of 3 network switches and 3 network routers, providing practical experience. Housed under shared desks, two pods connect to 14U racks, each with 12 patch ports for Ethernet and console connections. These racks contain essential network security equipment such as Firewalls, routers, Net Taps, and Wireless Access Points, facilitating collaborative learning. Course delivery follows the University Academic Calendar, with timetabled sessions during teaching weeks. Students are expected to dedicate three to four hours per week per 15-credit module for lectures, tutorials, and laboratories. Connect and Reflect Weeks, interspersed throughout each semester, allow students to consolidate learning, reflect on progress, set development goals, and focus on learning priorities. The course offers flexibility through themed pathways by elective choice, introduced in the common foundation year and expanded upon from Stages 2 to 4. During Stages 2 and 3, students study two core Computing modules and four subject modules core to the cyber security. Additionally, they elect to study a subject core module from a different course within the School of Computing to gain a broader understanding of the context in which Computing operates and diversify their skill set. The Placement serves as a valuable learning tool, offering an experience challenging to replicate through other means. Students benefit from interaction with company staff, exposure to required tasks, and immersion in the professional environment. Support during the Placement is provided by a course tutor from the School's staff and an industrial tutor from the company, ensuring a comprehensive learning experience with monitoring of the assessment process.
UCAS Code	Code I100
Course Collaborations	
No collaborations listed	
Introduction to the Course	Our BSc in Cyber Security equips students with essential knowledge and skills to navigate the ever-changing landscape of digital security. In response to the increasing interconnectedness of our world, the program focuses on securing sensitive information and safeguarding digital assets. This four-year course addresses the demand for professionals capable of effectively managing and securing digital technologies. Graduates are sought after for their ability to develop robust software systems, assess network security, and implement protection measures using various technologies and management processes. The curriculum covers diverse modules in cyber security, including network security, cryptography, ethical hacking, digital forensics, risk management, and compliance. Through hands-on experiences, students engage in simulating real-world cyber attacks, conducting forensic investigations, and designing secure architectures, fostering practical expertise tailored for the complexities of the modern cyber landscape. Beyond meeting industry standards, the course offers students the flexibility to choose from several specialized topics aligned with the School of Computing's active research areas. These encompass data science, artificial intelligence, cyber security, creative design, and game design, providing a unique opportunity for students to delve into areas of specific interest and relevance within the dynamic field of computing. In alignment with RGU's commitment to sustainability, the course strategically addresses sustainability goals by securing digital environments, mitigating cyber threats, and reducing their ecological impact. Educating professionals in Cyber Security practices reinforces trust in digital technologies and promotes eco-friendly practices, supporting sustainability objectives. Led by experienced faculty, students gain up-to-date insights into the latest cyber threats and defense mechanisms. Enrolling in our program is a journey that sharpens technical abilities, nurtures critical thinking, and instills ethical decision-making. Upon completion, graduates emerge well-prepared for roles such as Cyber Security analysts, ethical hackers, security consultants, or risk management specialists. They make significant contributions to the constantly evolving Cyber Security landscape, forging a rewarding career in one of the most vital and sought-after fields of the 21st century.

Overall Aim of the Course

Graduates of the BSc (Hons) Cyber Security degree course will attain:

- A systematic comprehension of fundamental computer science elements, specifically emphasizing cyber security, providing them with a comprehensive skill set in areas such as ethical hacking, network security, and incident response.
- An understanding of the threats confronting modern information infrastructures and the ability to deploy appropriate countermeasures to defend against such threats. The program actively integrates theoretical knowledge with practical application through hands-on experiences, simulated scenarios, and placement opportunities, ensuring graduates are not only well-informed but also proficient in applying their skills in real-world cyber security challenges.
- The capability to offer guidance to organizations on establishing secure systems, demonstrating an understanding of the inherent uncertainty, ambiguity, and trade-offs in the process. The curriculum is designed to keep students abreast of current cyber security threats, technologies, and best practices, ensuring graduates are well-prepared for the dynamic landscape of the cyber security industry.
- Proficiency in using established analytical and investigative techniques within the computing discipline to develop methods for testing an information system's security. The degree program aims to foster critical thinking and problem-solving skills, enabling students to analyze complex cybersecurity issues, identify vulnerabilities, and formulate effective strategies to safeguard digital assets.
- Recognition of the legal, social, ethical, and professional issues involved in cyber security, guided by the adoption of industry standards and professional practices. A key objective is to instill a strong sense of ethics and responsibility in students, emphasizing the importance of conducting cyber security activities within legal and ethical frameworks. Graduates are expected to be well-versed in the principles of responsible hacking and ethical conduct in the digital realm.

The course generally aligns with recognized benchmarks such as the QAA Computing benchmark statement, The Cyber Security Body Of Knowledge (CyBOK), the ACM Computer Science curriculum, and the Cyber Security Principles and Learning Outcomes for Computer Science and IT-Related Degrees published by (ISC)2. Beyond the technical focus, the curriculum includes content that prepares students for the social, ethical, legal, and professional aspects of a collaborative human environment, such as the workplace.

Aims of Each Stage of the Course

Stage 1 (Certificate of Higher Education)

The students will undertake a common Foundation Year, to allow them to understand the breadth of Computing as a multidisciplinary subject, with a particular focus within their chosen course. They will demonstrate proficiency in essential technical concepts, including software development, computer networking, web design, database systems, and principles of cyber security, incorporating hands-on application of relevant tools and techniques. The student will also acquire skills, knowledge, and understanding that promote continued personal and professional development through investigative, modeling, and problem-solving activities that use a combination of written and computational work, both on an individual and group basis. Additionally, students will have the opportunity to select two core modules from different courses within the School of Computing, providing a broader perspective on the operational context of Computing.

Stage 2 (Diploma of Higher Education)

The students will advance their comprehension and proficiency in aspects of computing critical to Cyber Security. This encompasses delving into areas such as Social & Human Factors in Cyber Security, Operating Systems & Virtualisation Security, Securing Networks, and Cryptography. Beyond technical expertise, emphasis is placed on honing research and professional skills pertinent to entrepreneurship and collaboration. An integral aspect of this stage involves providing students with the opportunity to showcase their ability to apply technology to address real-world problems, considering legal, ethical, and social implications. Additionally, students have the flexibility to select two modules from distinct courses within

Educational Aims of the Course

the School of Computing. This tailored approach allows them to construct a well-rounded skill set and knowledge base, incorporating elements from computing, creative design, or game development. This personalized learning path is designed to align with their preferred career trajectory, ensuring a comprehensive and adaptable skill portfolio.

Stage 3 (BSc Cyber Security)

The students will further advance their skills and knowledge by evaluating advanced methods in Information Risk & Security Management, Web & Mobile Security, Software Security & Malware Analysis, and Ethical Hacking. They will actively engage in collaborative efforts, assuming a leading role in planning and controlling computer security applications, both independently and as effective members of small project teams. The aim is to develop and assess computer systems proficiently, employing a diverse range of processes and techniques. Students will leverage various tools and techniques to protect information systems, ensuring a thoughtful balance between utility and usability. Furthermore, they will design and implement security solutions tailored to a variety of digital technologies, demonstrating advanced proficiency in addressing contemporary challenges within the field of Cyber Security. Additionally, students will have the opportunity to further refine their skills and knowledge by selecting two modules from different courses within the School of Computing, aligning their learning with personal preferences and industry demand.

Stage 4 (BSc (Hons) Cyber Security)

The students aim to integrate practical research and design activities, anchoring the development and maintenance of secure information systems by considering users, tasks, and work context. This involves the thoughtful selection of tools and technologies, along with the planning, organization, and implementation of a computing project that considers resource constraints, human factors, client requirements, and professional standards. Additionally, students will assess policies and technologies to ensure the timely identification of information security threats and effective recovery from cyber attacks. They will apply their knowledge of secure information systems, threats, and vulnerability exploitation to evaluate the security of computer networks and mitigate associated risks. To personalize their academic journey, students will choose a module from a complementary computing course that aligns with their earlier selections. Throughout the second semester, students will undertake a capstone project, utilizing a suitable combination of tools, technologies, and techniques from at least one academic discipline. This project will involve planning, designing, implementing, and evaluating a novel solution or research artifact for a substantial and unstructured problem, with a focus on user experience, human factors, client requirements, and professional standards.

Awarding Institution

Robert Gordon University

Course Duration

Full-Time - 4 years with a September intake or 5 years with the optional one year industrial placement.

Distinctive Features of the Course	The BSc Cyber Security course stands out with features tailored for a comprehensive and cutting-edge education. The curriculum covers foundational principles to advanced techniques, prioritizing hands-on learning through real-world projects, simulations, and lab exercises. Utilizing two dedicated cyber security labs in the school ensures that students remain at the forefront of industry trends. In the first year, all undergraduate courses in the School of Computing run as a foundation year, concentrating on the degree subject. This structure enables students to comprehend 'computing' as a multidisciplinary subject, fostering a community of practice across the School. As they progress, students delve into more advanced areas in cyber security, gaining comprehensive knowledge in all core cyber security knowledge areas. Consequently, graduates are well-equipped to develop secure software applications, maintain network security, monitor information systems for vulnerabilities, assure against cyber threats, and, if necessary, develop robust procedures for information recovery. Another distinctive feature is the alignment of course content with industry standards and sector guidelines. Providing a core cyber security skill set, the course enables students to progress to various industry-standard qualifications such as Certified Ethical Hacker (CEH), CCNA Security, and Certified Wireless Network Analyst (CWNA), enhancing employability and career prospects. The optional industrial placement offers students the opportunity to gain invaluable experience working with computer and network security professionals in a business environment. Completion of an industrial placement awards an additional credited module based on a review of the student's work-related skills by the industrial supervisor and visiting tutor. Taught by a team of full-time staff with extensive experience in the field, the course allows students to benefit from their expertise in applying security measures to information systems in business applications. The course adopts a practical approach to developing professional skills through group work and project teams, fostering personal development in leadership, management, teamwork, and communication, ultimately enhancing employability. Furthermore, the course places a strong emphasis on ethical and legal aspects, ensuring responsible conduct in the ever-changing digital landscape.
Placement/Professional Experience	Students may elect to undertake a short summer placement in a summer break and/or a year-long placement between stages. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who the student can contact about any academic or non-academic problems during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short summer placement between Stages 1 and 2, or Stages 2 and 3, or Stages 3 and 4. This normally comprises at least 4 weeks of continuous, full-time supervised work experience. The assessment for a short summer placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short summer placement will normally receive at least 1 video call with their Academic Tutor. In addition, or as an alternative to a short summer placement, students may undertake a year-long placement either between Stage 2 and 3, or between Stage 3 and 4. The year-long industrial placement normally comprises of 30 - 48 weeks of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or

	through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 120 credits achievable from taught modules at each stage. Successful completion is not required for progression to the following stage. The credits achieved for placement may not be used to compensate for any credits from the taught modules.
Opportunities for Further Study	An Honours degree achieved with at least an upper second class (Class 2.1) Classification will normally provide access to postgraduate and doctoral programmes in UK higher education institutions.
External and Internal Indicators of Quality and Standards	SCQF credit values have been utilized as a guiding framework during the design of modules, with consideration given to the Practical/Engineering Applications criteria set by the British Computer Society (BCS). Enriched within our core structure are modules covering essential computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques, Project Management, and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's program, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's undergraduate program. In addition to QAA guidelines, insights are drawn from the Cyber Security Body Of Knowledge (CyBOK), a comprehensive resource designed to guide and support education and professional training in the cyber security sector. Access to the CyBOK knowledge base can be gained at https://www.cybok.org/knowledgebase1_1/. Additionally, the International Information System Security Certification Consortium, Inc., (ISC)² guidelines and ACM Computer Science Curriculum 2013 (Curriculum Guidelines for Undergraduate Degree Programs in Computer Science) were referenced. Internally, the School employs robust processes and procedures to uphold and enhance standards. Stakeholder input on standards is actively sought through mechanisms like the Staff /Student Liaison Committee, Undergraduate Programme Management Team, Student Experience questionnaires, Industrial Board, and industrial visits to placement students. Our commitment to best practices in teaching and learning is reflected in active participation in University quality enhancement and assurance processes, including the regular five-yearly cycle of Institution-Led Subject Reviews and involvement in University quality assurance committee structures. Pedagogical research is undertaken by lecturers within the School, with a commitment to expanding the scholarship of computing science education. The School's dedication to academic excellence is underscored by its participation in university conferences on teaching and learning.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure

	Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Stage 1 Entry SQA Higher: BBCC and two further passes to include English and Maths at National 5 grade C or above if not held at Higher. GCE A Level Year 1 entry: CCC and two further passes to include English and Maths at GCSE grade 5 if not held at A Level. Irish Leaving Certificate Higher: H3H3H3H3 and two further passes to include English and Maths at grade O3 or above if not held at Higher. IB Diploma Year 1 entry: 26 to include a pass in Standard Level English and Maths at grade 4 or above if not held at Higher Level. Advanced entry into Stages 2 and 3 joining either the 'Continuing' or 'Articulation' routes will be subject to conditional/fixed elective modules or similar depending on subject area and this will form part of the course induction process
Alternative Qualifications for Entry	Equivalent qualifications, e.g SQA, BTEC, S/NVQs, overseas qualifications and access programmes will be considered on an individual basis. Applications from mature students with non-standard qualifications or work experience are welcomed.
Interview Procedures	Prospective applicants for the course are not normally interviewed.
Course-Specific Regulations	Honours Classification is based solely on the student's performance in Stage 4 of the course. CE4321 Capstone Project cannot be condoned [refer Academic Regulation A4, paragraph 11]. The credits achieved for placement are in addition to the 120 credits achievable from taught modules, and may not be used to compensate for any credits from the taught modules. Electives will run at the discretion of the School. Students cannot choose an elective module that they have already taken on the course, for example, some electives are aimed at particular cohorts such as articulating students. Some electives will not be available until the module is first delivered by the School, therefore, please consult the Course Leader for further information on elective availability.
	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The

Equality, Diversity and Inclusion	University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.

Philosophy of Course Approach to Learning

There are two main objectives to the BSc (Hons) Cyber Security degree. The first is to provide students with the knowledge and understanding of security issues in relation to the design, development and use of digital information systems. The second is to give students the ability to recognise the legal, social, ethical and professional issues involved in the exploitation of computer technology and be guided by the adoption of appropriate professional, ethical and legal practices.

The course will concentrate on five themes:

- information and risk,
- threats and attacks,
- cyber security architecture and operations,
- secure systems and products,
- cyber security management

These themes will all be underpinned with software development, networking and professional skills.

Information and Risk :

The pervasive use of digital technology to manage and control information resources means that the information infrastructures within all organisations are subject to threats by those who wish to exploit security vulnerabilities. Information risk assessment and management is concerned with the identification, estimation, evaluation and administration of the risk to critical information resources within an organisation, and the coordinated response to avoid or mitigate the harm that can be caused from the failure to properly manage or protect those assets. The management of risk allows organisations to identify vulnerabilities and threats to their information resources, and to decide their response to reduce risk to an acceptable level, based on the value of those resources. This can be done by minimising vulnerabilities, subjecting the threats to a process of prioritisation, with the deployment of appropriate countermeasures to protect assets from harm. Students will learn how the attributes relating to confidentiality, possession or control, integrity, authenticity, availability, and utility, can make information vulnerable to attack, as well as some of the legal and regulatory drivers, customer rights or organisation objectives which necessitate protection. In addition, they will develop an understanding of the concept of a risk landscape, its dynamic nature and how to create such a landscape for an organisation. They will learn how to classify threats and the concept of residual risk and what it means for an organisation. They will also evaluate the qualitative and quantitative criteria used to assess the suitability of risk management approaches for an organisation.

Threats and Attacks:

An understanding of the mechanisms by which information systems can be attacked, and the approaches that can be used to defend and protect against those attacks, is central to maintaining a secure and reliable information infrastructure. The core concepts in this area are that of a threat and the vulnerability it exploits, as well as the idea of an attack vector on information and the systems that process, store and transmit that information, and the typical targets of these attacks. Human factors involved such as social engineering, user behaviour and awareness and the mind-sets of hackers are also important. By the end of the course, students should be able to understand the difference between terms such as threat, risk, attack and vulnerability, describe how threats materialise into attacks, typical threats, attacks and exploits, and the motivations behind them, as well as where to find information about these things. Types of malware (e.g. viruses, Trojans and spyware), their method of distribution and their detailed mechanism of operation will also be covered as well as more advanced

Course-Specific Learning Experience

topics, such as the concept of a threat landscape and how to go about creating one for an organisation.

Cyber Security Architecture and Operations:

The protection of information, services and systems relies on a range of technical and procedural activities, often grouped in a framework. This information architecture will contain a variety of controls that can be implemented across an organisation to reduce information and systems risk, identify and mitigate vulnerability, and satisfy compliance obligations. Students will learn how to highlight the need for security architecture and describe its relevance to systems, service continuity and reliability. They will discuss the application of techniques such as defence in depth to demonstrate how controls can be selected, deployed and tested to minimise risk and impact, and differentiate between controls that protect systems availability and reliability, those that protect information and those that manage human behaviour.

Secure Systems and Products:

This theme looks at the concepts of design, defensive programming and testing and their application to build robust, resilient systems that are fit for purpose. One key concept is that of the software product lifecycle, with its various stages such as requirements gathering, design, build and testing. A good understanding of requirements analysis and other operational factors such as the need to balance levels of availability with an appropriate risk appetite, fault tolerance, load balancing, and the use of particular attack vectors are all needed. Strategic testing techniques such as use and misuse testing and defensive programming will be covered. The student should be able to understand the concepts that must be considered in requirements analysis and explain the importance of embedding security throughout the lifecycle, (i.e. "building security in"). This requires an understanding of the key steps in a design process and where security considerations should be implemented, especially physical, operational, logical and technical controls to protect systems and information. In addition, defensive programming techniques will be covered as well as strategic testing mechanisms.

Cyber Security Management:

The final main theme looks at the relationship between security considerations, the requirements of the organisation, the legal and regulatory environments in which it operates and to the efforts and abilities of individuals in handling information correctly. The core concepts concern the accountability and expectations of employees, and the organisational policies and strategies that promote compliance. Security becomes an ethical issue within IT practice and the assessment, analysis and management of risk, and the balance between opportunity and security, and the trade-offs required to achieve that balance, is related to the organisational culture in which information is managed and protected. The course should allow students to demonstrate a knowledge of a management framework and identify commonly used standards and areas of best practice, as well as the ability to place security in an organisational context related to respect for organisational needs, other individuals, and sensitive and confidential information. Students should be able to demonstrate an understanding of compliance and its importance and describe the various tools that can be used in cyber security management. This awareness of ethical concerns is a key to providing a positive security influence and is a significant factor when considering the creation of an awareness of security issues on a user education programme.

Technical Facilities

Many of the practical aspects of the course take place in the computer laboratory. The School of Computing Science and Digital Media has 9 laboratories housing a total of 335 PCs and 24 Macs, equipped with the latest versions of open source and commercial software for the exclusive use of Computing students.

The school has established a Cyber Security Lab (N519/N523) equipped with a dedicated setup for each student, featuring a CCNA/CCNP equivalent of equipment, including 3 network switches and 3 network routers known as a "Pod." Two pods are housed in a 14U rack beneath a desk shared by two students. Each rack provides 12 patch ports for a combination of Ethernet connectivity and console connectors per pod. Of these, 6 ports are utilized to connect to local devices and consoles, while the remaining 6 are connected to two centralized main racks. These main racks house a subset of network security equipment, including a firewall, net taps, and a wireless LAN controller for shared use. The second rack incorporates a switch with a fiber connection wired back to the University firewall, ensuring controlled access to the general Internet for specific MAC addresses. This access is limited to a single virtual machine (VM) while maintaining the security of the main RGU network. Students log in to this PC normally, accessing university resources like any other IT lab. The only configuration difference is the addition of 3 Virtual Machines on the PC, accessible through VMware Workstation. Each PC is upgraded with an internal 1TB hard disk and 3 additional network cards occupying the remaining slots in an HP Small Form Factor machine.

In addition to the Cyber Security Lab, students in the course have access to a networking lab (N424). This lab comprises 36 PCs connected to racks of Cisco equipment, representing three separate locations (Red, Green, and Blue). Each PC is connected back to these racks with three Local Area cables and a console cable. Students access university resources on

these PCs as they would in any other IT lab, with the only configuration difference being the addition of 3 Virtual Machines on the PC accessed through VMware Workstation. Each PC is equipped with 3 extra network cards, filling the remaining slots in an HP Small Form Factor machine. This lab mirrors the Cyber Security Lab installed in N519/N523, contributing to an enhanced teaching and pedagogical environment, along with an increased overall capacity.

Throughout the School there is 100% wireless coverage that allows access to the University network for any student who owns a wireless-enabled laptop.

The School's labs are regularly updated on a 3 year replacement cycle to ensure that the School's computing facilities are capable of delivering high performance and can run the latest software.

Access to the labs is by electronic key fob at any time during University opening hours.

Software provision is comprehensive, including compilers for most modern programming languages, database systems including Oracle, SQL server and MS Office packages. On the School's PCs, standard installed software includes (amongst others) Microsoft Office Professional, Adobe Creative and Production suites, Java, SPSS and Minitab.

The Learning Process: Teaching Strategy and Methods

The course curriculum is designed as a coherent whole with an emphasis on the integration of theory, practical skills and applications. It enables the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment and supplies methodologies for the adoption of a disciplined approach to digital media development tasks such as problem definition, specification, design, implementation, and maintenance. Such knowledge includes an understanding of the range of possible options for these tasks, suitable technologies and media types, the related practical and transferable skills including relevant approaches to collaborative work, access to the appropriate software tools and resources. It also provides the necessary underpinning to guide practice, to ensure the sustainability of their knowledge and to provide an appropriate framework that will accommodate rapid technological change.

These skills are developed at the degree and honours levels corresponding to the Scottish Credit and Qualifications Framework levels 9 and 10. As the student progresses through the course they are encouraged to expand their understanding and critical appreciation of both computing, networking and security technologies, and to extend skills in the analysis, design, implementation, testing and management of information security solutions.

A general feature of the course is that it stresses a unified, integrated view of the subject area. This is reflected in the fact that it is designed to:

- Present a coherent underpinning of the theory appropriate to the aims of the course which is further developed and used throughout the period of study. This enables graduating students to adapt to future developments in the field.
- Reflect the rapid rate of change in the field and ensure that coverage is given to a selection of emerging topics so that students are aware of likely future developments in the subject together with their potential impact.

- Integrate the delivery of teaching across thematic boundaries to highlight fundamental aspects of interconnectivity and interdisciplinary work within different areas of the syllabus.
- Integrate modes of assessment to attempt to balance the types (individual and group) and purposes (e.g. formative and summative) of such activity.
- Facilitate students to exhibit competence and achievement across a number of different themes (integrated assessment).
- Encourage students to combine elements of their learning from different parts of the programme and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment).

The professionally-orientated nature of the course requires students to obtain general and specific skills, this being addressed by a combination of teaching and learning methods, including lectures, tutorials, computing laboratories, workshops, seminars, practical work, industrial placement (where appropriate), integrative real-world project work, student-centred learning and private study.

Teaching and Learning Approaches

The main goal of the teaching and learning strategy is to enable students to become critical, reflective, independent learners who are active participants in the learning process and who develop the motivation and autonomy needed to take control and regulate their own educational experiences. This means that, throughout their course of study, students will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their degree, they will be in a position to evaluate and assess their own effectiveness.

This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide-range of appropriate delivery styles. These include conventional instructional teaching methodologies, social learning activities such as group and team working, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning. As well as the day-to-day use of the latest computer hardware and software afforded by the degree, extensive use of technology-assisted learning is made within the programme of study. Examples of these include assessment by means of computer-assisted assessment packages embedded in the virtual learning environment to engage the student and increase the speed of feedback and the use of an integrated approach whereby students have short periods of instruction followed by immediate lab exercises to contextualise the topic.

Modes of Delivery

Lecture

This is usually a formal discourse for the purposes of dissemination of information, the demonstration of techniques where appropriate and the discussion of supporting ideas and consequences. The lecture is supported by a network of dissemination equipment including computer projection facilities, whiteboard, OHP, and video where appropriate. Although this type of presentation is suitable for a one-sided discourse ample opportunity exists for questions, interaction and discussion and this is facilitated by technological means, e. g. through the use of personal response systems.

Tutorials

Within the School of Computing, this format is used for a wide range of activities, each suited to the particular subject. It may consist of the staff supporting students engaged in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities, allocate tasks, etc. Generally this type of teaching is used to support the lecture, clarify the material and experiment with the techniques and skills required.

Laboratory

The nature of the computing elements of all courses requires students to gain practical skills in the use of a variety of computer technology. This activity takes place in one of the School's computer laboratories and consists of the student, supported by staff members, practising skills in the use of sophisticated software applications and including software development and

systems analysis and design tools. There has been increasing emphasis on using a “language–lab” approach to subjects in which a series of short mini-lectures are delivered in labs to allow students to gain theoretical knowledge of a topic. These are interspersed with selected practical exercises, which allow this knowledge to be immediately assimilated and so reinforce the learning process.

Industrial Experience

The course includes an Industrial Placement, which is a useful supplement to the other teaching and learning activities of the course. Students participating in the Industrial Placement have to competitively apply for placements by submitting letters of application and CVs and by attending interviews. Students on placement undertake a variety of roles and tasks for the company they are working for, supported on the Placement by a course tutor from the staff of the School and an industrial tutor from the company. The interaction with the staff of the company, the required tasks and the general environment provides an experience difficult to achieve by other means. The School tutor supports and monitors the assessment process.

In addition to the Industrial Placement, there is a series of presentations and seminars given by academic staff, staff from the careers advisory service and professionals working in key positions within the Computing and Information Technology sectors of business and industry. These presentations and seminars form part of the core Professional Development module.

Group Project

Group projects involve extensive collaborative efforts typically carried out under the guidance of a faculty member or mentors within a team-oriented module environment. Students are grouped into small project teams and provided with a task specification that may encompass aspects such as analysis, design, and implementation of information systems, software development, or the exploration and practical investigation of related activities. The students autonomously manage the project processes, functioning as a cohesive team to generate a set of deliverables. These deliverables often include a design and development component featuring interactive systems artifacts. Additionally, teams are generally expected to produce accompanying documentation to meet professional standards. This activity closely mirrors the development of projects undertaken by industry development teams, offering students an authentic experience that aligns with a diverse range of industries. Through participation in the team-based module, students hone skills essential for real-world applications.

Capstone Project

This is an extended piece of work, completed under supervision of a staff tutor. The students agree on a specification for a task, which may be the analysis and design of a system, the development of software, or similar, and produce a finished thesis and product. This activity replicates the development process that would occur in an industrial development or research setting.

Student Role in Learning

Students are expected to be active participants in the learning process. This means that there is a requirement for students to extend their learning from formally delivered materials through wider reading, research and self-directed enquiry. As students develop over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their own personal objectives for learning. Students are encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which may form the basis of tutorial guidance with staff. Students are expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library.

The Scottish Credit and Qualifications Framework (SCQF) provides a description of the learning experience in terms of five broad areas, as well as descriptors for each level. These can be found in the relevant sections of the Core Academic Data for this course. However, the main learning objectives for the course, in terms of the five characteristics of Knowledge and Understanding, Practice: Applied Knowledge, Skills and Understanding, Generic Cognitive Skills, Communication, ICT and Numeracy Skills, and skills to promote Autonomy,

Accountability and Working with Others, are given below.

Characteristic 1: Knowledge and Understanding

The student should have knowledge and understanding of:

- The principles and application of the concepts involved in the specification, design, implementation, testing and maintenance of secure software systems.
- The principles and application of the concepts involved in the application of computer and network security practices.
- The role of digital information systems within an organisation, the promotion of judgement and decision-making capabilities in evaluating risk associated with solutions to the cyber security problems of organisations and individual users, and their specification and implementation where necessary.
- The issues involved in the management and security of digital information and computer technology in an organisation, and the development of secure industrial information systems.

Characteristic 2: Practice: Applied Knowledge, Skills and Understanding

The applied knowledge, skills and understanding necessary to:

Teaching and Learning Strategy

- Analyse the requirements of users and organisations, to design, implement, evaluate and manage secure information systems.
- Use appropriate tools to evaluate the security of information systems in terms of general quality attributes and possible trade-offs presented within a given problem and context.
- Deploy effectively the tools used for the construction and documentation of secure computer applications, with particular emphasis on understanding the whole process involved in the effective deployment of software to solve practical problems.
- Work as an effective member of a project team, managing the interface between computing, the organisation and the end-users, recognising the different roles within a team and different ways of organising teams.
- Select an appropriate range of tools and technologies and plan, organise and implement a cyber security project with due regard to human factors, user and organisational requirements.
- Carry out practical research assisting organisations to deploy effective information security technologies.
- Effectively communicate complex ideas and technical issues using verbal, visual and written techniques and media appropriate to the target audience.

Characteristic 3: Generic Cognitive Skills

The student should have the general cognitive skills to:

- Analyse, evaluate, and interpret complex information and use such knowledge and understanding in the modelling, design and implementation of software solutions to individual user and organisational problems.
- Recognise and analyse criteria and specifications appropriate to specific problems, and plan strategies for their solution.
- Analyse and evaluate the extent to which an information system meets the criteria defined for its current level of security and predicted future requirements.
- Select and deploy appropriate theory, practices and tools for the specification, design, implementation and evaluation of digital information systems.
- Research and assess the use of new technologies and develop imaginative and practical solutions to a diverse range of problems.
- Research, gather and collate data from a range of sources, establish relevant connections, and apply to a given context to inform the generation of design proposals.
- Plan and manage work appropriately, and to work to deadlines.
- Develop an understanding of personal learning needs and methods.

Characteristic 4: Communication, ICT and Numeracy Skills

The student should be able to:

- Use effective means of communication to express complex ideas about cyber security through appropriate media, based on considerations of context.
- Make effective use of specialist ICT facilities concerned with the management of data and information, develop appropriate skills for this purpose, and adjust features to suit purpose.
- Demonstrate knowledge and understanding through investigative, modelling and problem solving activities.
- Present, formally and informally, information about specialised cyber security topics to informed audiences.
- Communicate with peers, senior colleagues and specialists on a professional level.
- Select, interpret, use and evaluate a wide range of numerical and graphical data to set and achieve cyber security goals, and to use this information as evidence when presenting arguments for security decisions.

Characteristic 5: Autonomy, Accountability and Working with Others

The student should be able to:

- Adopt a broad-ranging and flexible approach to study, identifying learning needs, pursuing activities designed to meet these needs in increasingly autonomous ways.
- Recognise the need for lifelong learning and undertake appropriate continuing professional development, taking control of the learning process including organisational skills, time and workload management.
- Work effectively both independently and as a member of a team, showing a willingness to learn from others.
- Maintain professional standards and quality control as applied to the specification, production, implementation and maintenance of business software and information systems.
- Be innovative, foster creativity and drive, and manage the process of change.

	The Assessment of Student Work The University has developed a Grading Scheme for assessment purposes, details of which may be accessed at: www.rgu.ac.uk/academicregulations . It is important to note that all grades indicated to students during the course of the academic session are provisional until the deliberations of the Assessment Board for the course. Provisional grades are presented to this Board, which consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each Module undertaken by them. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal. Any potential barriers to students with disabilities or specific needs are assessed by the School's Disability Coordinator who liaises with appropriate teaching staff to ensure that such students are not disadvantaged during an assessment. In the case of examinations, the University has a specific policy which must be followed: Policy: Assessment and Students with Special Needs • Students with special needs must register by a specified deadline date to have their special examination requirements considered. Approval shall not be given for special examination arrangements unless the student was registered in advance. • Recording of a student's special needs examination requirements shall be on a standard proforma, which should be agreed by the student and reviewed for each academic session. • Where appropriate, provision of further information on students' special needs shall be recorded on the examination timetable; for example, mobility difficulties that might impact on the choice of examination venue. The Assessment of Student Work A variety of assessment methods are used in this course, as deemed appropriate to the subject material being assessed and the context in which the material is delivered. Work-Based Learning The optional Industrial Placement is assessed by a combination of Tutor and Supervisor assessment based on a form of Learning Contract supported by a Learning Log. Computer Aided Assessment Computer-assisted assessment packages integrated into the virtual learning environment such as Moodle's question system are used in many modules, both for formative and summative assessment, to enhance student engagement and increase the speed of feedback. Peer Assessment Peer Assessment is utilised in many modules throughout the course where group work is undertaken. This serves to develop working and interpersonal skills, as well as honing appraisal

Assessment

and critique skills. Peer assessment may be used in both summative and formative contexts.

Unseen Exams

In general unseen exams are used to assess students' knowledge and understanding, and as the students progress through the course unseen exams become a larger component of the assessment of taught modules.

Assessed Laboratories

Assessed Laboratories are used to assess students' knowledge, understanding and practical skills, and generally involve students undertaking an unseen practical task such as developing a part of an information system, under closed book, supervised conditions with a fixed time constraint.

Coursework

For summative assessment the achievement of Learning Outcomes is assessed by a combination of exam and coursework. Coursework may take many forms, i.e. reports, essays, production of software, design documentation and modelling activities. The School Office prepares a schedule of coursework assessments for each teaching semester in order to balance the students' workload.

Projects/Dissertations

Extended projects are used in later stages as a learning vehicle and assessment process that allow students to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. These projects often require the practical application of a number of different technologies and provide students with the opportunity to learn new investigative and evaluative skills. Projects, in particular group projects, provide an opportunity to learn new investigative and evaluative skills, and are also often used as the vehicle for gaining valuable employability skills. Project work is usually assessed by a combination of assessment of the completed project deliverable as an artefact, and an accompanying report or dissertation documenting the project work undertaken by the student.

Assessment Types

Many modules use a combination of assessment types that integrate modes of assessment in order to balance the types of activity (e.g. individual and group activities) and purposes of assessment (e.g. formative and summative).

Where there are a number of modules that cover allied subject material integrated assessment may be used to facilitate students to exhibit competence and achievement across a number of different themes.

In all years, synoptic assessments embedded into projects and other practical assignments are used to encourage students to combine elements of their learning from different parts of the programme and to show their accumulated knowledge and understanding of a set of topics or a subject area.

A student's Honours Classification (First Class Honours, Upper Second Class Honours etc) is based solely on performance in Stage 4 of the course.

Equality and Diversity

Any datasets used within the course will aim to adhere to the Software Engineering Code of Ethics and Professional Practice recommended by the ACM/IEEE-CS Joint Task Force which states that "only accurate data derived by ethical and lawful means" will be used, and that its integrity will be maintained. Furthermore, students will be taught about recognising and researching known bias within datasets, and where possible, will be signposted to use datasets that are less likely to have inherent bias.

University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>; • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	In addition to the University-wide support for Teaching and Learning, the School provides a number of means of supporting students in teaching and learning. Members of staff publish course based teaching and learning materials on the University's Virtual Learning Environment, Moodle. The range of published materials spans from supplementary teaching materials such as additional lecture notes and tutorial guides to software development to general guidance for students, such as codes of conduct in laboratories. Guidelines for course elements such as the Honours Project module can also be found here. In addition to this, students can access study support material from the Department for the Enhancement of Learning, Teaching and Access (DELTA). This is the central department that supports the Schools in implementing the university's strategic goals in teaching and learning, extending access and the student experience. http://www.rgu.ac.uk/staff-and-current-students/staff-area/delta/department-for-the-enhancement-of-learning-teaching-and-access/ Course materials such as lecture notes in the form of PDF documents are available on the School's website for some of the courses. Because of licensing restrictions on some materials, some of the links are for internal viewing only. Where course materials require the use of particular files, these are made available on the shared drive (for internal viewing only). They can be found in folders under S:\Computing\staff-admin\teaching. Information on studying abroad can be accessed at http://www.rgu.ac.uk/future-students/study-abroad Feedback is provided continuously to students through informal contact with subject lecturers and tutors in the tutorial and laboratory settings. Feedback is provided on assessed practical work normally within three weeks of submission of the work. Facilities are made available for all students to view their marked examination scripts. Grades are made available electronically for viewing via the student portal following ratification by the Assessment Board.

University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	Annual Course/Programme Appraisals are reviewed and approved by the Programme Management Team and School Academic Board which meet at least once per semester. The Programme Management Team considers Teaching and Learning issues arising from feedback generated by External Examiner Annual Reports, Staff/Student Liaison Committees, and Student Experience Questionnaires, and these issues are further dealt with where appropriate by the School's Teaching Committee, chaired by the School's Learning Enhancement Coordinator. The Programme Management Team and School Academic Board also review all proposed changes to the structure and content of the Undergraduate Computing Programme. The Undergraduate Computing Programme is also informed by feedback from the School's Industrial Board and the Faculty's Quality Enhancement Sub-Committee.
	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements.

University Student Feedback Mechanisms	This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	There is a single Staff/Student Liaison Committee for all undergraduate degree courses. It normally consists of the Head of School, Undergraduate Programme Leader and Course Leaders, with student representatives from each stage of each course. The Committee meets at least twice per semester and feeds appropriate information to other departments and/or committees as required. Student representatives also take part in the School Academic Board where they can contribute to discussions about School policy. In addition, there are regular scheduled meetings between Course Leaders and student representatives to discuss issues arising within specific classes and on individual modules. Module Evaluation Questionnaires are distributed to students on all modules early in the second half of the semester which allows them to raise issues and give feedback to staff

Awards

Stage 1

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Certificate of Higher Education - CertHE	Computing	Aberdeen, United Kingdom	Exit Award		120

Stage 2

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Diploma of Higher Education - DipHE	Computing	Aberdeen, United Kingdom	Exit Award		240

Stage 3

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Bachelor of Science - BSc	Cyber Security	Aberdeen, United Kingdom	Exit Award		360

Stage 4

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Bachelor of Science with Honours - BSc (Hons)	Cyber Security	Aberdeen, United Kingdom	Target Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	The principles and application of the concepts involved in the specification, design, implementation, testing and maintenance of secure software systems.	SC1	
LLO2	The principles and application of the concepts involved in the application of computer and network security practices.	SC1	
LLO3	The role of digital information systems within an organisation, the promotion of judgement and decision-making capabilities in evaluating risk associated with solutions to the cyber security problems of organisations and individual users, and their specification and implementation where necessary.	SC1	
LLO4	The issues involved in the management and security of digital information and computer technology in an organisation, and the development of secure industrial information systems.	SC1	
LLO5	Analyse the requirements of users and organisations, to design, implement, evaluate and manage secure information systems.	SC2	
LLO6	Use appropriate tools to evaluate the security of information systems in terms of general quality attributes and possible trade-offs presented within a given problem and context.	SC2	
LLO7	Deploy effectively the tools used for the construction and documentation of secure computer applications, with particular emphasis on understanding the whole process involved in the effective deployment of software to solve practical problems.	SC2	
LLO8	Work as an effective member of a project team, managing the interface between computing, the organisation and the end-users, recognising the different roles within a team and different ways of organising teams.	SC2	
LLO9	Select an appropriate range of tools and technologies and plan, organise and implement a cyber security project with due regard to human factors, user and organisational requirements.	SC2	
LLO10	Carry out practical research assisting organisations to deploy effective information security technologies.	SC2	
LLO11	Effectively communicate complex ideas and technical issues using verbal, visual and written techniques and media appropriate to the target audience.	SC2	
LLO12	Analyse, evaluate, and interpret complex information and use such knowledge and understanding in the modelling, design and implementation of software solutions to individual user and organisational problems.	SC3	
LLO13	Recognise and analyse criteria and specifications appropriate to specific problems, and plan strategies for their solution.	SC3	
LLO14	Analyse and evaluate the extent to which an information system meets the criteria defined for its current level of security and predicted future requirements.	SC3	
LLO15	Select and deploy appropriate theory, practices and tools for the specification, design, implementation and evaluation of digital information systems.	SC3	
LLO16	Research and assess the use of new technologies and develop imaginative and practical solutions to a diverse range of problems.	SC3	
LLO17	Research, gather and collate data from a range of sources, establish relevant connections, and apply to a given context to inform the generation of design proposals.	SC3	
LLO18	Plan and manage work appropriately, and to work to deadlines.	SC3	
LLO19	Develop an understanding of personal learning needs and methods.	SC3	

LLO20	Use effective means of communication to express complex ideas about cyber security through appropriate media, based on considerations of context.	SC4	
LLO21	Make effective use of specialist ICT facilities concerned with the management of data and information, develop appropriate skills for this purpose, and adjust features to suit purpose.	SC4	
LLO22	Demonstrate knowledge and understanding through investigative, modelling and problem solving activities.	SC4	
LLO23	Present, formally and informally, information about specialised cyber security topics to informed audiences.	SC4	
LLO24	Communicate with peers, senior colleagues and specialists on a professional level.	SC4	
LLO25	Select, interpret, use and evaluate a wide range of numerical and graphical data to set and achieve cyber security goals, and to use this information as evidence when presenting arguments for security decisions.	SC4	
LLO26	Adopt a broad-ranging and flexible approach to study, identifying learning needs, pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	
LLO27	Recognise the need for lifelong learning and undertake appropriate continuing professional development, taking control of the learning process including organisational skills, time and workload management.	SC5	
LLO28	Work effectively both independently and as a member of a team, showing a willingness to learn from others.	SC5	
LLO29	Maintain professional standards and quality control as applied to the specification, production, implementation and maintenance of business software and information systems.	SC5	
LLO30	Be innovative, foster creativity and drive, and manage the process of change.	SC5	
LLO31	Describe a basic knowledge and understanding of the basic context of computer science, and its relationship to other subject areas such as mathematics, science and engineering.	SC1	SCQF 7
LLO32	Outline a familiarity with the basic facts and principles of computing and supporting topics via demonstrations of skills gained, the ability to answer questions and to describe and employ basic tools and techniques. The student is introduced to the basic skills and knowledge required for operating in a computer environment via study and practice. The student should understand, at an appropriate, elementary level, programming and the development of software, computer hardware and network design, and show an understanding of the business environment and roles played by information systems in organisations.	SC1	SCQF 7
LLO33	Recognise the importance of cultivating a practical understanding of fundamental facts and principles in computing and information security through the study of reports, solutions to standard problems, and projects that focus on core fundamentals, demonstrating an appropriate use of conventions as outlined in the topics above.	SC1	SCQF 7
LLO34	Carry out study of the subject in a systematic, directed way with the aid of appropriate tutor guidance within the classroom and laboratory.	SC2	SCQF 7, SCQF 7
LLO35	Apply knowledge, skills, and understanding of computer science, fundamentals of cybersecurity, and computer networks in practical contexts.	SC2	SCQF 7
LLO36	Employ some of the basic and routine professional skills, techniques, and practices associated with computing.	SC2	SCQF 7
LLO37	Apply these skills in both routine and non-routine contexts.	SC2	SCQF 7
LLO38	Apply the basic principles and procedures found in computer science and cyber security to standard, simple situations.	SC3	SCQF 7
LLO39	Present the ability to analyse situations and problems, based on the set of classifications presented by tutors.	SC3	SCQF 7
LLO40	Systematically relate to a limited number of facts/ideas/elements in an imitative manner.	SC3	SCQF 7
LLO41	Start to define their own value judgments based on the criteria provided.	SC3	SCQF 7

LLO42	Apply the basic tools/methods of computing and cyber security to simple, standard problems	SC5	SCQF 7
LLO43	Convey aspects of computer science in a clear and concise way, in writing and orally, in relatively informal and limited-length pieces of work. The student should also be able to communicate appropriately the basic design and documentation of programs.	SC4	SCQF 7
LLO44	Achieve basic numeracy, mathematics and the ability to manipulate data at a level appropriate to this stage of the degree. The student should also develop an understanding of basic concepts in logic and computation.	SC4	SCQF 7
LLO45	Obtain practical skills in computer science using basic ICT applications as well as those skills necessary for efficient use of computer networks.	SC4	SCQF 7
LLO46	Manage learning in an increasingly effective and purposeful way, showing the beginnings of development as an autonomous learner, and exercise some initiative and independence in carrying out defined activities at a professional level of practice.	SC5	SCQF 7
LLO47	Carry out interactions effectively with tutors and fellow students via software and web development tasks, problem-solving tutorials and the production of appropriate documentation, and accept supervision in less familiar areas of work.	SC5	SCQF 7
LLO48	Exercise some managerial or supervisory responsibility for the work of others within a defined and supervised structure while managing limited resources within defined areas of work.	SC5	SCQF 7
LLO49	Take the lead in implementing agreed plans in familiar or defined contexts, while taking account of your own and others' roles and responsibilities when carrying out and evaluating tasks.	SC5	SCQF 7
LLO50	Work, under guidance, with others to acquire an understanding of current professional practice.	SC5	SCQF 7
LLO51	Report an understanding of the important central areas of this stage of the degree, including the integration of professional skills, human factors in cyber security, secure system operations, research methodologies, network security, and cryptography into a unified and comprehensive framework.	SC1	SCQF 8
LLO52	Infer a coherent picture of the discipline by furthering their appreciation of the scope of the field of study.	SC1	SCQF 8
LLO53	Compare the significance of central and peripheral areas of the subject, suggesting opportunities to explore and extend the boundaries of the discipline.	SC1	SCQF 8
LLO54	Practice knowledge, skills, and understanding in using a range of professional skills, techniques, and practices associated with the computing topics studied, some of which are at an advanced or complex level.	SC2	SCQF 8
LLO55	Show the ability to carry out routine lines of inquiry, development, or investigation into professional-level problems and issues.	SC2	SCQF 8
LLO56	Adapt routine practices within accepted standards.	SC2	
LLO57	Use standard principles and procedures to somewhat more demanding situations.	SC3	SCQF 8
LLO58	Undertake the application of classifications and engage in critical analysis of relatively simple situations, showcasing a growing proficiency.	SC3	SCQF 8
LLO59	Undertake the task of recognizing the importance of connecting and organizing a variety of facts, ideas, and elements within a reasoned argument.	SC3	SCQF 8
LLO60	Compare a range of valid alternative solutions, and begin to discriminate and evaluate in a reasoned, systematic and increasingly independent way, particularly within the context of group work.	SC3	SCQF 8
LLO61	Use given tools/methods accurately and carefully to more demanding problems appropriate to this stage.	SC3	SCQF 8
LLO62	Use communication skills in more formal academic and professional styles, and in more extended pieces of work. Some presentation work may be introduced through the optional placement process.	SC4	SCQF 8
	Undertake the application of a selection of more advanced numerical, modelling and design skills, relevant to specialised areas of professional		

LLO63	skills, human factors in cyber security, secure system operations, research methodologies, network security, and cryptography.	SC4	SCQF 8
LLO64	Use a broad selection of more advanced computing and IT skills in a wide range of contexts in all topics, but particularly in the development of solutions to computer security problems.	SC4	SCQF 8
LLO65	Practice autonomy and initiative by comparing some activities at a professional level within the range of exercises set by tutors.	SC5	SCQF 8
LLO66	Interact effectively within a learning- or subject-specific group, demonstrating basic negotiating, role-ownership, leadership, and group-support skills.	SC5	SCQF 8
LLO67	Plan and take the lead on work activities in familiar or defined contexts and conclude them successfully.	SC5	SCQF 8
LLO68	Take responsibility for managing resources and the work of others within a group, within a defined structure.	SC5	SCQF 8
LLO69	Practice in ways that show awareness of your own and others' roles, responsibilities, and contributions when undertaking and evaluating tasks.	SC5	SCQF 8
LLO70	Work, under guidance, with others to compare and acquire an understanding of current legal, ethical, and professional issues, and manage these issues in accordance with current professional and/or ethical codes of practices.	SC5	SCQF 8
LLO71	Discuss important central areas of this stage of the degree; Information Risk & Security Management, Ethical Hacking, Web & Mobile Security, Software Security & Malware Analysis.	SC1	SCQF 9
LLO72	Explain the range of the principles, main theories, concepts, and terminology of information security.	SC1	SCQF 9
LLO73	Discuss the broad areas of the knowledge and subject bases of computer science and information systems, including key specialist areas.	SC1	SCQF 9
LLO74	Explain the limits of the broad area of study, together with an ability to relate the different areas to one another in a systematic and logical manner.	SC1	SCQF 9
LLO75	Demonstrate knowledge, skills and understanding in using a range of the principal professional skills, techniques, practices and/or materials associated with cyber security.	SC2	SCQF 9
LLO76	Demonstrate applied knowledge of some specialised or advanced skills, techniques, practices and/or materials in the design, implementation, communication, and marketing of cyber security solutions.	SC2	SCQF 9
LLO77	Demonstrate application of routine methods of enquiry and research.	SC2	SCQF 9
LLO78	Demonstrate ability to work in a range of professional contexts that include a degree of unpredictability.	SC2	SCQF 9
LLO79	Demonstrate the ability to select and apply principles and procedures relevant to the current situation or problem.	SC3	SCQF 9
LLO80	Recognize familiar ideas or principles in new contexts or situations and systematically and effectively analyze them.	SC3	SCQF 9
LLO81	Compile facts, ideas, and elements in an articulated case or design, especially in the context of developing cyber security solutions across various situations.	SC3	SCQF 9
LLO82	Differentiate among a range of valid alternative solutions; begin to discriminate and evaluate in a reasoned, systematic, and increasingly independent way using a range of sources.	SC3	SCQF 9
LLO83	Self-direct, identify key elements of problems, and choose appropriate methods to construct their solution, with only general guidance, across all topics.	SC3	SCQF 9
LLO84	Effectively interpret communication requirements of a variety of roles, to produce clear, well-structured reports and presentations on relevant aspects of cyber security.	SC4	SCQF 9
	Defend the utilization of specific numerical, modeling, and design skills essential for the development and implementation of secure systems		

LLO85	and networks.	SC4	SCQF 9
LLO86	Access and successfully use a broad selection of specialist computing and ICT skills in a wide range of contexts, in all topics.	SC4	SCQF 9
LLO87	Demonstrate a broad-ranging and flexible approach to study, identifying learning needs and pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	SCQF 9
LLO88	Assume an appropriate role in the design and management of secure information systems development projects.	SC5	SCQF 9
LLO89	Demonstrate the ability to plan, oversee, and be accountable for the installation and maintenance of secure software systems.	SC5	SCQF 9
LLO90	Demonstrate adept negotiation and collaboration skills in managing the interface between computing, information security, and other functional areas within an organization, as well as with users of its computer systems.	SC5	SCQF 9
LLO91	Assess and determine appropriate responses to legal, ethical, and professional issues in accordance with current codes of best practice, seeking guidance when necessary.	SC5	SCQF 9
LLO92	Illustrate an understanding of the context, relationship, and interplay among the principal areas, features, boundaries, terminology, and conventions within the specialized domains of the course. These specialized areas at this level encompass Digital Forensics & Analysis, Security Operations & Incident Management, and AI & Cyber Security.	SC1	SCQF 10
LLO93	Illustrate confident familiarity and critical understanding of the defining concepts and features of secure Information Systems, including further areas of special relevance or interest and show a detailed understanding of the ways in which the subject is developed, including a range of established techniques of enquiry or research methodologies.	SC1	SCQF 10
LLO94	Develop appropriate high-level expertise in relevant implementation skills, user experience, and artificial intelligence techniques, and apply specialist analysis within the Capstone Project, while learning technical and design skills appropriate to the specific task.	SC1	SCQF 10
LLO95	Communicate detailed knowledge and understanding in one or more specialisms related to cyber security, some of which are informed by, or at the forefront of the subject.	SC1	SCQF 10
LLO96	Conceptualise a clear understanding of their subject's boundaries, the limits of its application, and of the range of methods of study and types of judgment employed.	SC5	SCQF 10
LLO97	Execute knowledge, skills and understanding in using a wide range of the principal professional skills, techniques, and practices associated with cyber security.	SC2	SCQF 10
LLO98	Illustrate practical skills in the use of techniques and materials that are specialised, advanced and at the forefront of the subject of information security.	SC2	SCQF 10
LLO99	Illustrate a practical knowledge of cyber security to execute a defined project of research, development or investigation in the Capstone Project module.	SC2	SCQF 10
LLO100	Operate in a range of professional-level contexts that include a greater degree of unpredictability and specialism in information security.	SC2	SCQF 10
LLO101	Compose an accurate selection and application of principles and procedures to the solution of a range of professional situations and problems, working autonomously, particularly within the Capstone Project module.	SC3	SCQF 10
LLO102	Examine principles and ideas in new contexts and situations, and analyse them systematically, effectively and critically, particularly within the Capstone Project module.	SC3	SCQF 10
LLO103	Compose facts, ideas and elements in support of a well-structured argument, design novel solutions and evolve new concepts, working autonomously, particularly within the Capstone Project module.	SC3	SCQF 10

LLO104	Reconcile theory with professional and vocational practice, autonomously evaluate theory, process, solutions and outcomes critically and effectively, particularly within the project.	SC3	SCQF 10
LLO105	Develop creative solutions to complex problems in an increasingly independent, confident and flexible way and apply knowledge and skills appropriate to their solution, at the threshold of professional competence in all topics, but, particularly within the Capstone Project module.	SC3	SCQF 10
LLO106	Justify the use of a broad selection of specialist computing and cyber security skills, in all topics, but particularly within the Capstone Project module.	SC4	SCQF 10
LLO107	Appraise the adept utilization of specialized computing and ICT facilities dedicated to the secure management of data and information, and cultivate requisite skills for this purpose.	SC4	SCQF 10
LLO108	Communicate effectively to express complex ideas to colleagues through appropriate media, based on considerations of context.	SC4	SCQF 10
LLO109	Engage effectively in independent roles, debate in a professional manner, and produce detailed critiques and coherent project reports. Students should be able to give confident oral and formal presentations to informed audiences in a wide range of contexts, particularly within the Capstone Project module.	SC4	SCQF 10
LLO110	Critique the demonstration of their skills, knowledge, and understanding through autonomous investigative, modeling, and problem-solving activities, and justify the process related to the development, maintenance, and review of secure information systems.	SC5	SCQF 10
LLO111	Reconcile standards and quality control in relation to the specification, production, implementation, and maintenance of secure information systems, aligning with current professional and ethical codes of practice.	SC5	SCQF 10
LLO112	Justify personal learning with minimal guidance, utilizing a broad range of resources appropriate to the profession, and making effective use of feedback, at this level within all topics, but particularly within the Capstone Project.	SC5	SCQF 10
LLO113	Argue effectively within groups on all topics and, where applicable, justify interaction and negotiating skills with customers and other stakeholders.	SC5	SCQF 10

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9

SC3	Generic cognitive skills	SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Stage 1 / Semester 1**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1131	Cyber Security Fundamentals		Harsha Kalutarage (hk9491)	Yes	2.0	15	Level 7
Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	No	1.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives 1	Cyber Security - Electives Group 1		Ian Harris (ish7850)		1.0	15	

Stage 1 / Semester 2**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1132	Computing Network Fundamentals		Jean-Claude Golovine (jrrg5708)	Yes	2.0	15	Level 7
Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives 2	Cyber Security - Electives Group 2		Ian Harris (ish7850)		1.0	15	

Stage 2 / Semester 1**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2134	Operating Systems and Virtualisation Security	Routes A, B	Ian Harris (ish7850)	Yes	1.0	15	Level 8
Full Time On Campus	CM2112	Real World Projects and Professional Skills	Routes A, B	Ross McLean (rjm7342)	Yes	1.0	15	Level 8
Full Time On Campus	CM2133	Social and Human Factors in Cyber Security	Routes A, B	Christopher McDermott (cdm7692)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives 3	Cyber Security - Electives Group 3	Routes A, B	Ian Harris (ish7850)		3.0	15	

Stage 2 / Semester 2**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2136	Cryptography	Routes A, B	Muhammad Shadi Hajar	Yes	1.0	15	Level 8

Full Time On Campus	CM2124	Research Methods	Routes A, B	Daniel Gardner	Yes	2.0	15	Level 8
Full Time On Campus	CM2135	Securing Networks	Routes A, B	Christopher McDermott (cdm7692)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives 5	Cyber Security - Electives Group 5	Routes A, B	Ian Harris (ish7850)		3.0	15	

Stage 2 / Semester 3**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives 7	Cyber Security - Electives Group 7		Ian Harris (ish7850)		2.0	120	

Stage 3 / Semester 1**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM3146	Ethical Hacking	Routes A, B	Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 9
Full Time On Campus	CM3144	Information Risk and Security Management	Routes A, B	Hatem Ahriz (cmsa)	Yes	1.0	15	Level 9
Full Time On Campus	CM3130	Interdisciplinary Team Project	Routes A, B	Mark Zarb (mz7371)	Yes	1.0	15	Level 9

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives 8	Cyber Security - Electives Group 8	Routes A, B	Ian Harris (ish7850)		3.0	15	

Stage 3 / Semester 2**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM3141	Pirate Studies	Routes A, B	Ross McLean (rjm7342)	Yes	1.0	15	Level 9
Full Time On Campus	CM3148	Software Security and Malware Analysis	Routes A, B	Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 9
Full Time On Campus	CM3145	Web and Mobile Security	Routes A, B	Hatem Ahriz (cmsa)	Yes	1.0	15	Level 9
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives10	Cyber Security - Electives Group 10	Routes A, B	Ian Harris (ish7850)		3.0	15	

Stage 3 / Semester 3**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives 7	Cyber Security - Electives Group 7		Ian Harris (ish7850)		2.0	120	

Stage 4 / Semester 1**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM4143	AI and Cyber Security		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 10
Full Time On Campus	CM4141	Digital Forensics and Analysis		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 10
Full Time On Campus	CM4142	Security Operations & Incident Management		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 10
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	PS0531 - Electives12	Cyber Security - Electives Group 12		Ian Harris (ish7850)		3.0	15	

Stage 4 / Semester 2**Full Time On Campus**

Route A - Continuing. Route B - Articulating. Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	No	1.0	60	Level 10

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time On Campus	CM1131	Cyber Security Fundamentals		Harsha Kalutarage (hk9491)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	No	1.0	15	Level 7
1	Semester 1	Elective	Full Time On Campus	PS0531 - Electives 1	Cyber Security - Electives Group 1		Ian Harris (ish7850)		1.0	15	
1	Semester 2	Core	Full Time On Campus	CM1132	Computing Network Fundamentals		Jean-Claude Golovine (jrg5708)	Yes	2.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
1	Semester 2	Elective	Full Time On Campus	PS0531 - Electives 2	Cyber Security - Electives Group 2		Ian Harris (ish7850)		1.0	15	
2	Semester 1	Core	Full Time On	CM2134	Operating Systems and Virtualisation Security	Routes A, B	Ian Harris (ish7850)	Yes	1.0	15	Level 8

			Campus								
2	Semester 1	Core	Full Time On Campus	CM2112	Real World Projects and Professional Skills	Routes A, B	Ross McLean (rjm7342)	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CM2133	Social and Human Factors in Cyber Security	Routes A, B	Christopher McDermott (cdm7692)	Yes	1.0	15	Level 8
2	Semester 1	Elective	Full Time On Campus	PS0531 - Electives 3	Cyber Security - Electives Group 3	Routes A, B	Ian Harris (ish7850)		3.0	15	
2	Semester 2	Core	Full Time On Campus	CM2136	Cryptography	Routes A, B	Muhammad Shadi Hajar	Yes	1.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CM2124	Research Methods	Routes A, B	Daniel Gardner	Yes	2.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CM2135	Securing Networks	Routes A, B	Christopher McDermott (cdm7692)	Yes	1.0	15	Level 8
2	Semester 2	Elective	Full Time On Campus	PS0531 - Electives 5	Cyber Security - Electives Group 5	Routes A, B	Ian Harris (ish7850)		3.0	15	
2	Semester 3	Elective	Full Time On Campus	PS0531 - Electives 7	Cyber Security - Electives Group 7		Ian Harris (ish7850)		2.0	120	
3	Semester 1	Core	Full Time On Campus	CM3146	Ethical Hacking	Routes A, B	Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 9
			Full								

3	Semester 1	Core	Time On Campus	CM3144	Information Risk and Security Management	Routes A, B	Hatem Ahriz (cmsha)	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CM3130	Interdisciplinary Team Project	Routes A, B	Mark Zarb (mz7371)	Yes	1.0	15	Level 9
3	Semester 1	Elective	Full Time On Campus	PS0531 - Electives 8	Cyber Security - Electives Group 8	Routes A, B	Ian Harris (ish7850)		3.0	15	
3	Semester 2	Core	Full Time On Campus	CM3141	Pirate Studies	Routes A, B	Ross McLean (rjm7342)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3148	Software Security and Malware Analysis	Routes A, B	Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3145	Web and Mobile Security	Routes A, B	Hatem Ahriz (cmsha)	Yes	1.0	15	Level 9
3	Semester 2	Elective	Full Time On Campus	PS0531 - Electives10	Cyber Security - Electives Group 10	Routes A, B	Ian Harris (ish7850)		3.0	15	
3	Semester 3	Elective	Full Time On Campus	PS0531 - Electives 7	Cyber Security - Electives Group 7		Ian Harris (ish7850)		2.0	120	
4	Semester 1	Core	Full Time On Campus	CM4143	AI and Cyber Security		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CM4141	Digital Forensics and Analysis		Harsha Kalutarage	Yes	1.0	15	Level 10

			Campus				(hk9491)				
4	Semester 1	Core	Full Time On Campus	CM4142	Security Operations & Incident Management		Harsha Kalutarage (hk9491)	Yes	1.0	15	Level 10
4	Semester 1	Elective	Full Time On Campus	PS0531 - Electives12	Cyber Security - Electives Group 12		Ian Harris (ish7850)		3.0	15	
4	Semester 2	Core	Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	No	1.0	60	Level 10

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Sources of Further Information

#	URL and Description
1	https://www.cybok.org/knowledgebase1_1/ Several external sources were consulted in the process of curriculum development for the BSc (Hons) Cyber Security degree: Cyber Security Body Of Knowledge (CyBOK), a comprehensive resource designed to guide and support education and professional training in the cyber security sector. Access to the CyBOK knowledge base can be gained at https://www.cybok.org/knowledgebase1_1/. A significant source of information on the Cyber Security subject area and how it should be embedded within the field of Computer Science is the whitepaper, 'Cybersecurity Principles and Learning Outcomes for Computer Science and IT-Related Degrees', produced jointly by the UK Council of Professors and Heads of Computing and the International Information System Security Certification Consortium, Inc., (ISC)². This can be accessed https://cphcuk.files.wordpress.com/2015/06/j0028-isc2-white-paper-a4-v5-260515lr.pdf. The accreditation guidelines for the BCS, The Chartered Institute for IT, Guidelines on course accreditation, Information for universities and colleges, January 2020, can be found at https://www.bcs.org/media/9006/course-accreditation-guidelines.pdf and Academic Accreditation Guidelines, April 2022 can be found at https://www.bcs.org/media/1209/accreditation-guidelines.pdf. The Association for Computing Machinery (ACM) Guidelines on curricular elements for Computer Science (and related areas), Computer Science Curricula 2013, can be found at https://www.acm.org/binaries/content/assets/education/cs2013_web_final.pdf.